Xiaohan Zhang

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RESEARCH SUMMARY

Robotics, Machine Learning, and Computer Vision

The North Star for my research is to develop learning and planning algorithms that enable autonomous robots to perform service tasks in home environments. Here are topics (with representative first-author publications) I am interested in:

- Vision-language grounding for Task and Motion Planning [C2] [C6]
- Foundation models for long-horizon decision making [P1] [C3]
- Multi-sensory object perception and exploration [J1] [C7]

ACADEMIC BACKGROUND

State University of New York at Binghamton
Binghamton, NY
Ph.D. in Computer Science — Aug. 2020 – May 2024 (Expected)

- Thesis Topic: Symbol Grounding for Task and Motion Planning in Robotics
- Advisor: Shiqi Zhang
- Thesis Committee: Mohan Sridharan, Lijun Yin, Jayson Boubin

University of Texas at Austin
Austin, TX
Visiting Ph.D. Student — Jan. 2023 – Apr. 2023

- Evolutionary optimization for task planners [C2], LLMs for classical planning [P2]
- Advisor: Peter Stone
- Collaborators: Yuke Zhu, Roberto Martín-Martín, Joydeep Biswas

Renmin University of China
Beijing, China
B.E. in Computer Science — Sep. 2015 – May 2019

EMPLOYMENT

Fundamental AI Research (FAIR), Meta
Pittsburgh, PA
Research Scientist Intern — May. 2023 – Dec. 2023

- Vision-language models and 3D scene representations for planning and skill learning [P1] [C1]
- Intern Host: Christopher Paxton
- Collaborators: Yonatan Bisk, Sasha Sax, Aravind Rajeswaran

Brain Robotics, Google (now Google DeepMind)
Mountain View, CA
Student Researcher — Mar. 2022 - Aug. 2022

- Reinforcement learning for cleaning robots [C5]
- Intern Host: Montserrat Gonzalez
- Collaborators: Fei Xia, Jie Tan, Wenhao Yu

PREPRINTS

[P1] **Xiaohan Zhang***, Arjun Majumdar*, Anurag Ajay*, Pranav Putta, Sriram Yenamandra, Mikael Henaff, Sneha Silwal, Paul Mcvay, Oleksandr Maksymets,

Sergio Arnaud, Karmesh Yadav, Qiyang Li, Ben Newman, Mohit Sharma, Vincent Berges, Shiqi Zhang, Pulkit Agrawal, Yonatan Bisk, Dhruv Batra, Mrinal Kalakrishnan, Franziska Meier, Chris Paxton, Sasha Sax, and Aravind Rajeswaran. *(*Equal Contribution)*
OpenEQA: Embodied Question Answering in the Era of Foundation Models.
ArXiv, 2023

[P2] Bo Liu*, Yuqian Jiang*, **Xiaohan Zhang**, Qiang Liu, Shiqi Zhang, Joydeep Biswas, and Peter Stone. *(*Equal Contribution)*
LLM+P: Empowering Large Language Models with Optimal Planning Proficiency.
ArXiv, 2023

[P3] Yan Ding, Cheng Cui, **Xiaohan Zhang**, and Shiqi Zhang.
GLAD: Grounded Layered Autonomous Driving for Complex Service Tasks.
ArXiv, 2022

JOURNAL ARTICLES

[J1] **Xiaohan Zhang**, Saeid Amiri, Jivko Sinapov, Jesse Thomason, Peter Stone, and Shiqi Zhang.
Multimodal Embodied Attribute Learning by Robots for Object-Centric Action Policies.
Autonomous Robots, 2023

[J2] Yan Ding, **Xiaohan Zhang**, Saeid Amiri, Nieqing Cao, Hao Yang, Chad Esselink, and Shiqi Zhang.
Integrating Action Knowledge and LLMs for Task Planning and Situation Handling in Open Worlds.
Autonomous Robots, 2023

[J3] Yan Ding, **Xiaohan Zhang**, Xingyue Zhan, and Shiqi Zhang.
Learning to Ground Objects for Robot Task and Motion Planning.
*IEEE Robotics and Automation Letters (**RA-L**), 2022*

CONFERENCE & WORKSHOP PAPERS

[C1] Priyam Parashar, Vidhi Jain, **Xiaohan Zhang**, Jay Vakil, Sam Powers, Yonatan Bisk, and Chris Paxton.
SLAP: Spatial-Language Attention Policies.
*Conference on Robot Learning (**CoRL**), 2023*

[C2] **Xiaohan Zhang**, Yifeng Zhu, Yan Ding, Yuqian Jiang, Yuke Zhu, Peter Stone, and Shiqi Zhang.
Symbolic State Space Optimization for Long Horizon Mobile Manipulation Planning.
*IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**), 2023*

[C3] **Xiaohan Zhang***, Yan Ding*, Chris Paxton, and Shiqi Zhang. *(*Equal Contribution)*
Task and Motion Planning with Large Language Models for Object Rearrangement.
*IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**), 2023*

[C4] **Xiaohan Zhang**, Yan Ding, Saeid Amiri, Hao Yang, Andy Kaminski, Chad Esselink, and Shiqi Zhang.
Grounding Classical Task Planners via Vision-Language Models.
ICRA Workshop on Robot Execution Failures and Failure Management Strategies, 2023

[C5] Thomas Lew, Sumeet Singh, Mario Prats, Jeffrey Bingham, Jonathan Weisz, Benjie Holson, **Xiaohan Zhang**, Vikas Sindhwani, Yao Lu, Fei Xia, Peng Xu, Tingnan Zhang, Jie Tan, and Montserrat Gonzalez.
Robotic Table Wiping via Reinforcement Learning and Whole-body Trajectory Optimization.
*IEEE International Conference on Robotics and Automation (**ICRA**), 2023*

[C6] **Xiaohan Zhang**, Yifeng Zhu, Yan Ding, Yuke Zhu, Peter Stone, and Shiqi Zhang.
Visually Grounded Task and Motion Planning for Mobile Manipulation.
*IEEE International Conference on Robotics and Automation (**ICRA**), 2022*

[C7] **Xiaohan Zhang**, Jivko Sinapov, and Shiqi Zhang.
Planning Multimodal Exploratory Actions for Online Robot Attribute Learning.
*Robotics: Science and Systems (**RSS**), 2021*

[C8] Kishan Chandan, Jack Albertson, **Xiaohan Zhang**, Xiaoyang Zhang, Yao Liu, and Shiqi Zhang.
Learning to Guide Human Attention on Mobile Telepresence Robots with 360 Vision.
*IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**), 2021*

[C9] Yan Ding, **Xiaohan Zhang**, Xingyue Zhan, and Shiqi Zhang.
Task-Motion Planning for Safe and Efficient Urban Driving.
*IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**), 2020*

ACADEMIC SERVICES

- Program committee member: AAAI (2023, 2024), IJCAI (2024)
- Reviewer: RA-L (2022, 2023), ICRA (2024), IROS (2021, 2022, 2023)